FP-4-07-05-06-001-980B-2

ACSA Workshop
23/08/2024 Salerno

DRASS®

Cybersecurity in the Defense and Commercial Underwater Field

Safeguarding Critical Infrastructure and Sensitive Data

Marco Bellomo
www.drass.tech

DRASS, 1927

DRASS, founded in 1927, is today a world leader in the Subsea and Underwater Technologies and Systems

Midget and Compact Submarines, Swimmer Delivery Vehicles for Special Forces, Hyperbaric and Diving Systems, Deep Dive Simulators, Submarine Rescue Systems, Last Generation Diving Helmets

FACILITIES

DRASS ROMANIA,
TIMISOARA, ROMANIA

OFFICES	1000 sqm
STORES	1300 sqm
WORKSHOP	6000 sqm

DRASS
LIVORNO, ITALY

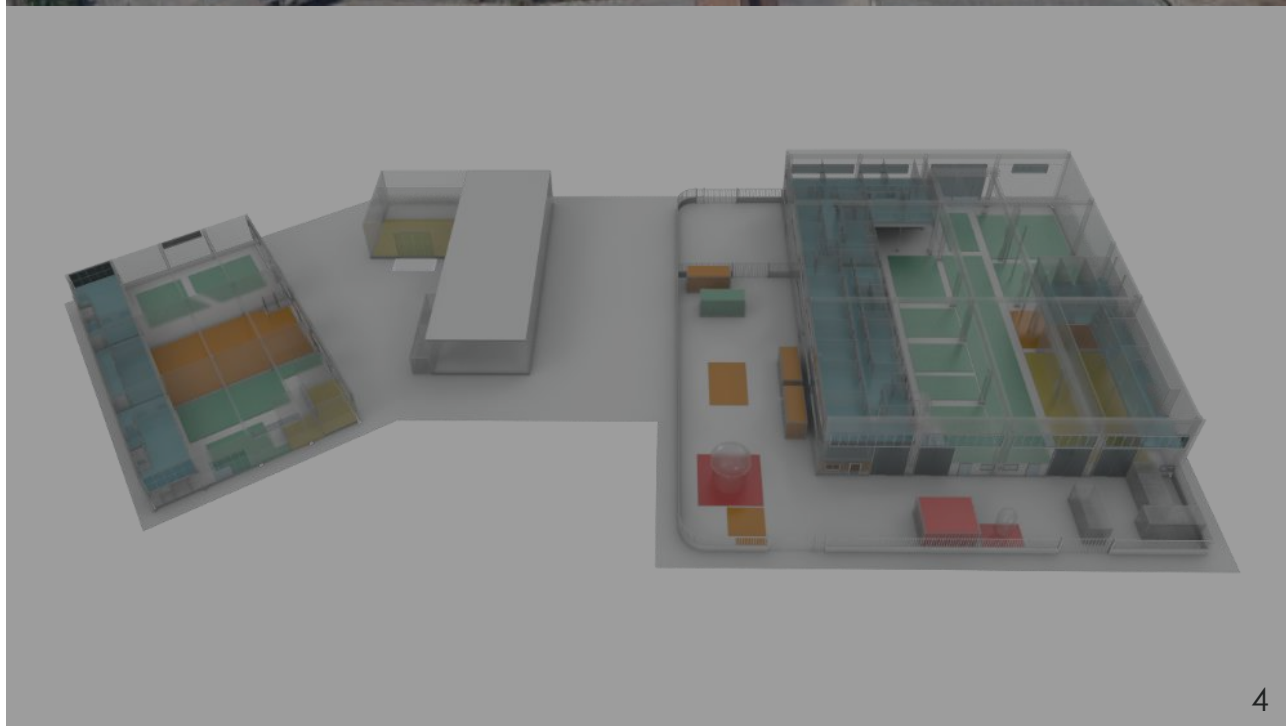
OFFICES	1300 sqm
STORES	1500 sqm
WORKSHOP	2400 sqm

DRASS MIDDLE EAST
DUBAI, UNITED ARAB EMIRATES

DRASS®

DRASS FACILITY, ITALY

- Automation, Electronic and Electrical Systems
- Control panels
- Outfitting of Diving Bells, Submarine Hulls
- Skids and Equipment
- Swimmer Delivery Vehicle
- High Technology Components



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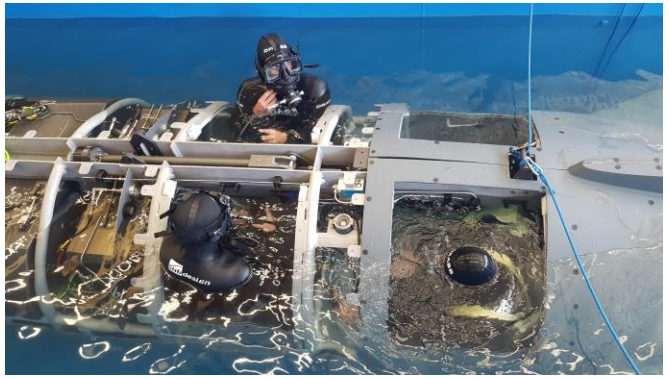
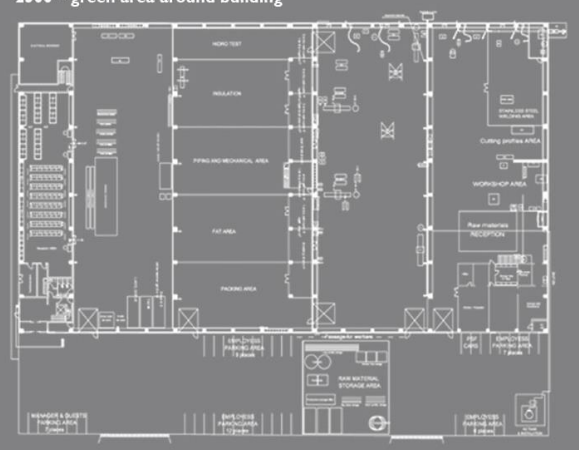
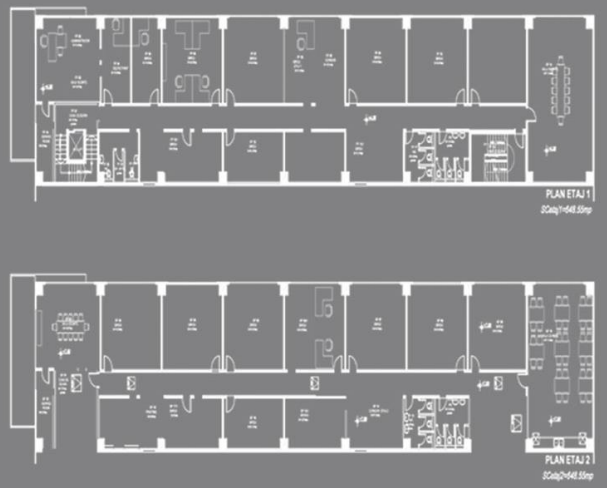


DRASS FACILITY, ROMANIA

- Submarine Heavy Carpentry & Foundations
- Complete Hyperbaric Chambers
- Metallic skids
- Gas control panels

Total surface mq: 12430
Storage Area mq: 720
Production Area mq: 5400
Office mq: 1610
Outer Area mq: 2200 – concrete platform
2500 – green area around building





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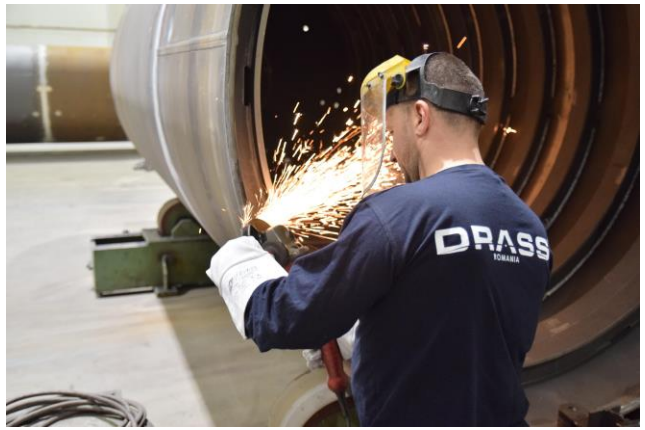
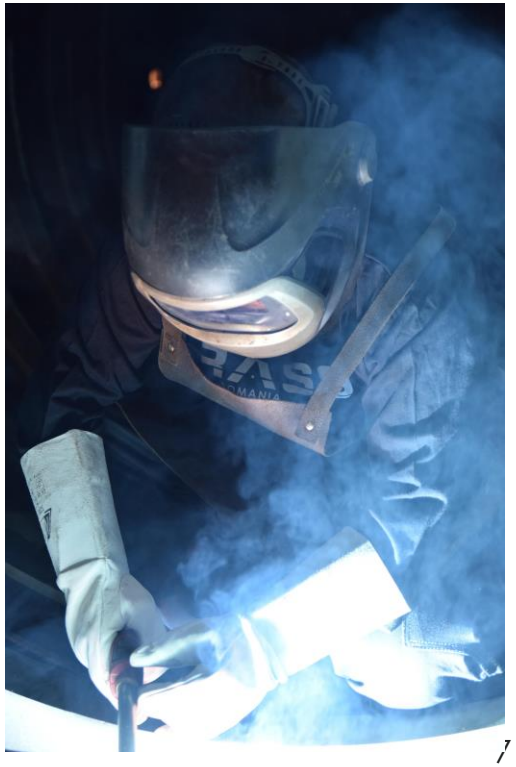




DRASS®

INDUSTRIALIZED PRODUCT

Serial production allows a comprehensive logistic support, documentation and reliability analysis.

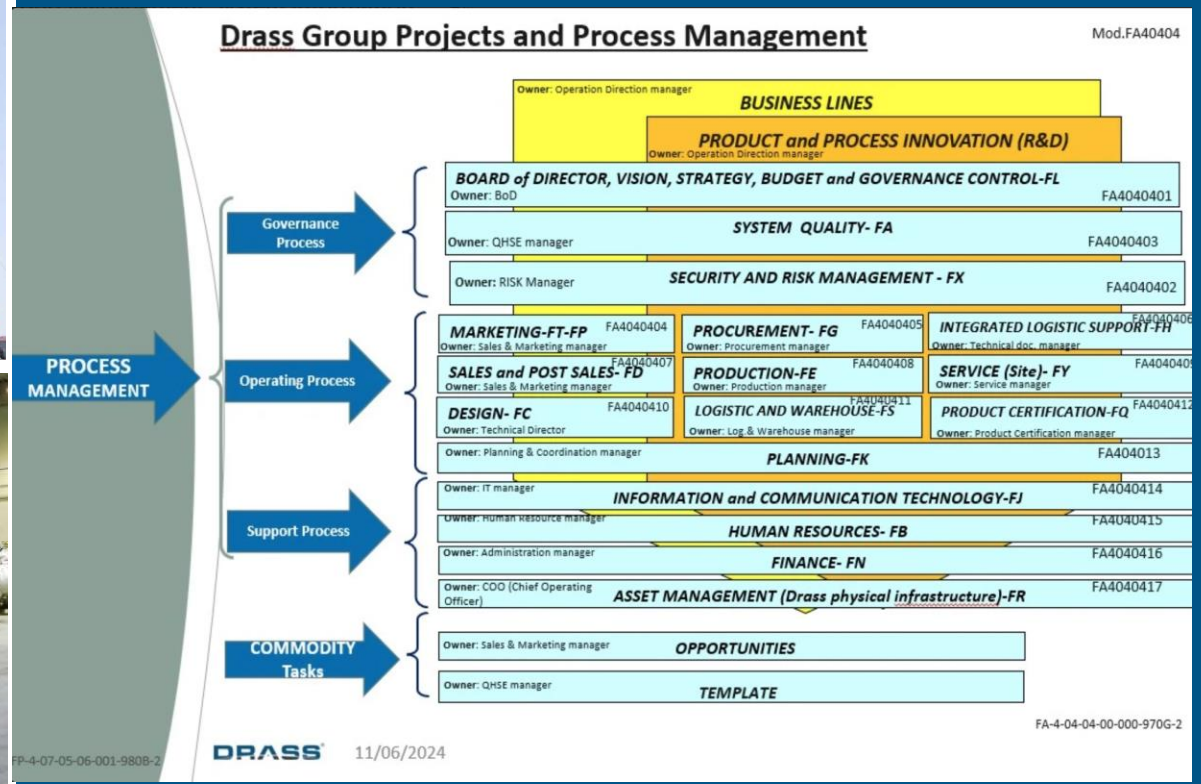
Feedback from Clients is looped back via technical bulletins to the existing installations in a continuous improvement cycle.

Well structured logistic support results in improved availability of spares and components.

DESIGN & LOGISTIC

The PPNN standardized and reduced, optimizing the logistic support. Different production lines use the same standardized components.

All PPNN/SSNN/Commercial Part Number are managed by an Integrated Logistic Design Support software suite developed by Drass and operating also with Partners and Clients.



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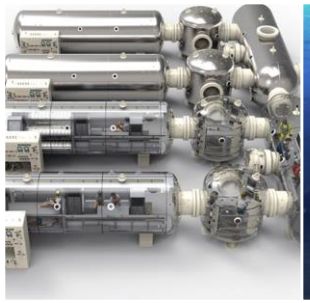
BUSINESS LINES OVERVIEW

DIVING & SIMULATOR

- Deep Diving
- Surface Diving
- Modular Systems



DEFENCE SOLUTIONS

- Midget Submarine
- Compact Submarine
- Swimmer Delivery Vehicle

SUBMARINE RESCUE & INTERVENTION SYSTEMS

- Resident Rescue System
- Deployable Rescue System
- Submarine Ventilation
- Submarine Intervention Systems

GLOBAL SERVICE

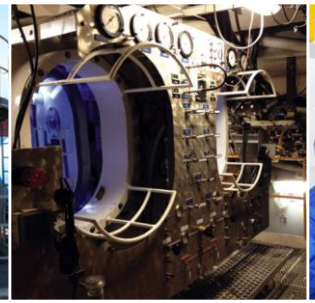
- Spares and Components
- Repair and Maintenance
- Training Centre

ENGINEERING & TESTING

- Consultancy
- Laboratory and Experimental

INDUSTRIAL

- Tunneling Hyperbaric Chambers
- Industrial Pressure Vessels



MEDICAL HOT SERIES & SCUBA

- Medical Hyperbaric Chambers
- Scuba Diving Chambers
- Diving Boats

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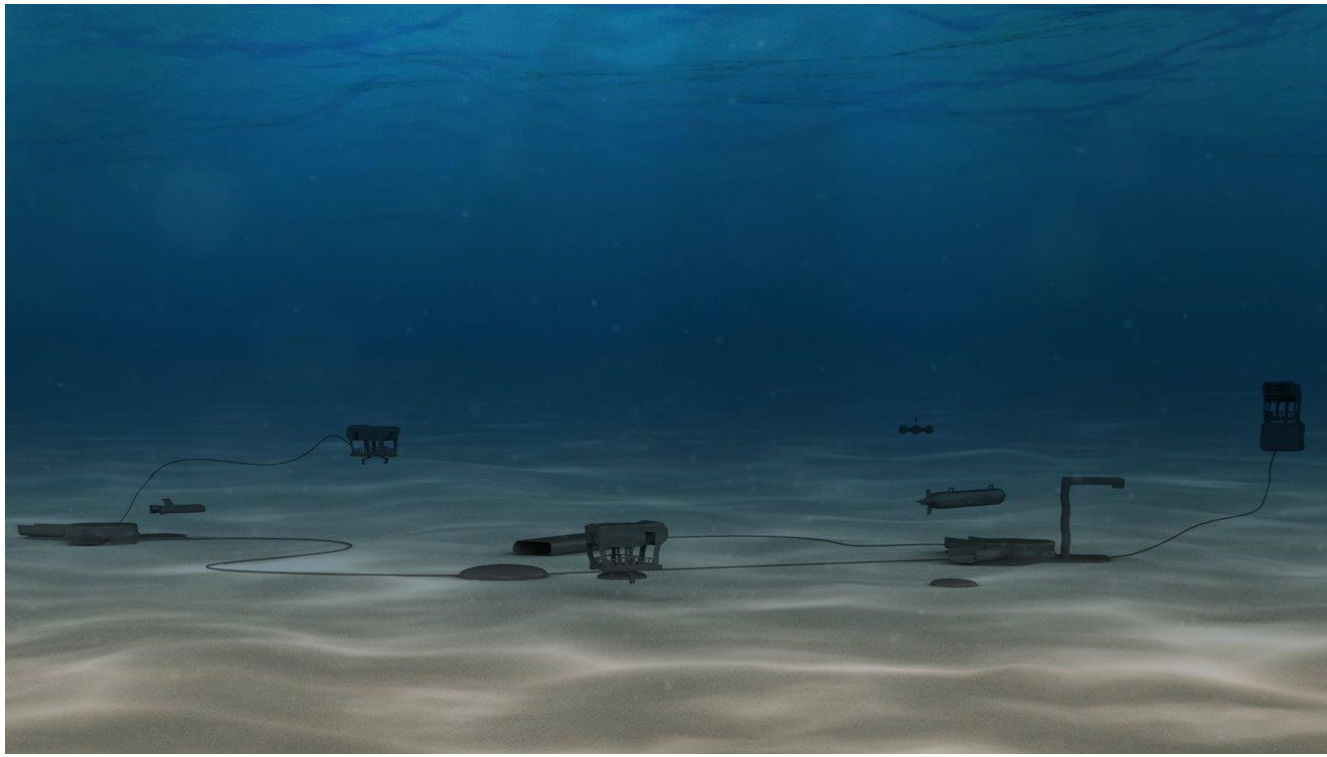
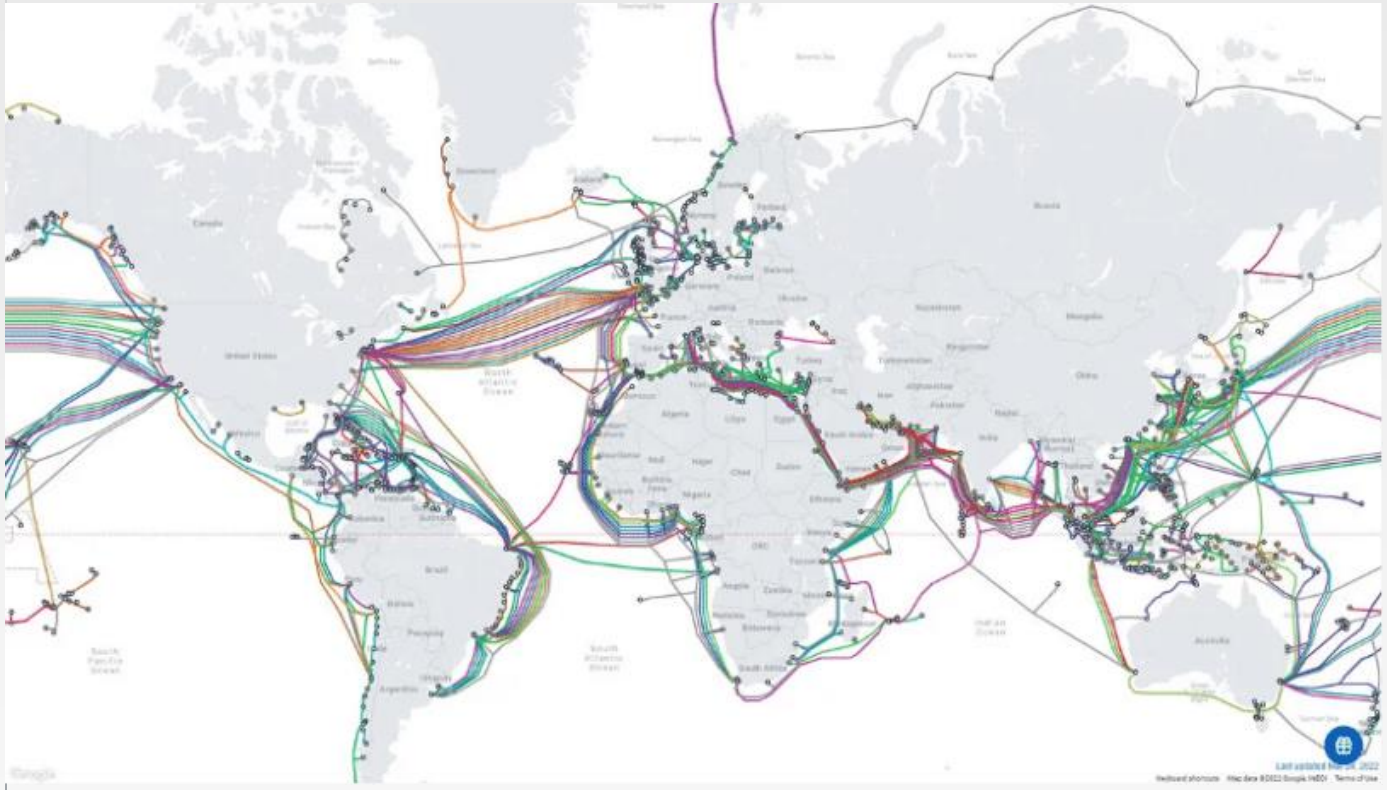
Why take care of Underwater Systems?

- Importance of underwater systems in defense and commercial sectors.
- Increasing reliance on underwater technology (submarines, underwater drones, communication cables, etc.).
- Growing cybersecurity threats targeting underwater infrastructure.



Folgiero (CEO of Fincantieri) also spoke about the underwater sector, whose market "**until 2030 will be around 400 billion euros"**. The Fincantieri group is “very focused on this new field. I think it's very similar to what space was like 40 years ago. So there is a lot of technology to validate. There are many new solutions and new needs to satisfy.
SHIPPING ITALY 20 FEBBRAIO 2024

Importance of Underwater Systems



Defense Sector:

- Submarines: Nuclear deterrence, intelligence gathering, and special operations.
- Unmanned Underwater Vehicles (UUVs): Surveillance, mine detection, and reconnaissance.

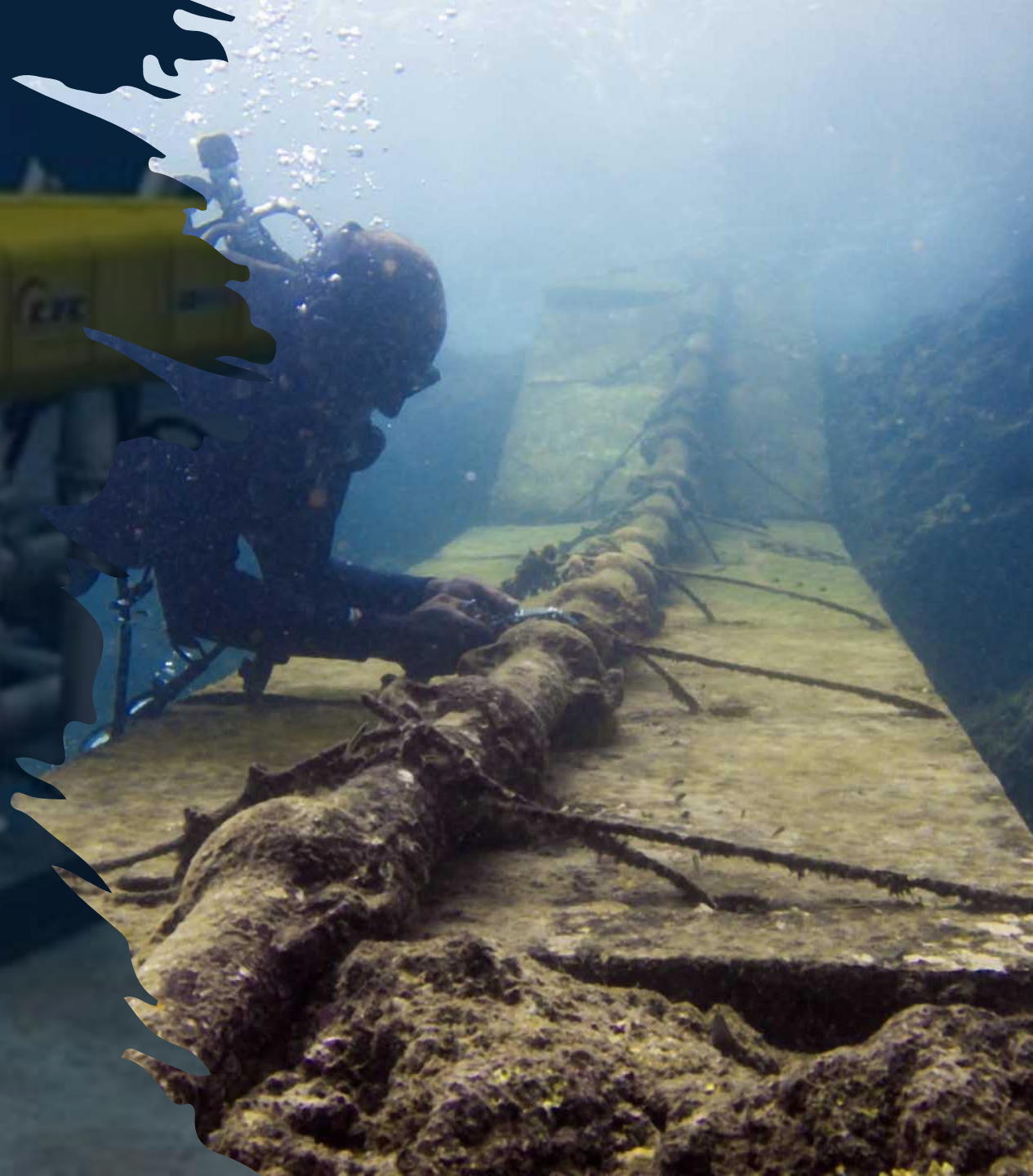
Commercial Sector:

- Subsea communication cables: Backbone of global internet connectivity.
- Offshore oil and gas platforms: Essential for energy supply.

Emerging Cybersecurity Threats

Types of Threats:

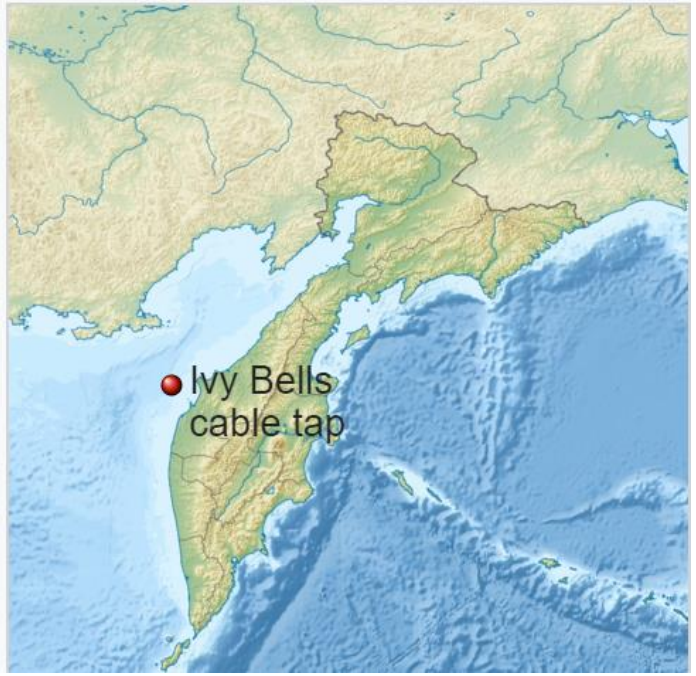
- Espionage: Targeting sensitive military communication and data transmission.
- Sabotage: Disrupting underwater infrastructure, including communication cables and energy platforms.
- Ransomware and Malware: Attacks on underwater systems to disrupt operations or steal data.
- Supply Chain Attacks: Compromise of components used in underwater systems.



Case Studies

Submarine Communication Interception

- **Real-world example of intercepted submarine communications.**
- **Impact on national security and military operations.**



Operation Ivy Bells was a joint United States Navy, Central Intelligence Agency (CIA), and National Security Agency (NSA) mission whose objective was to place wiretaps on Soviet underwater communication lines during the Cold War.

Red Sea cables have been damaged, disrupting internet traffic
LondonCNN — Mon March 4, 2024
Damage to submarine cables in the Red Sea is disrupting telecommunications networks and forcing providers to reroute as much as a quarter of traffic between Asia, Europe and the Middle East, including internet traffic.

Cyber-attack on submarine cables to date occurred in April 2022, when the US federal government revealed it had thwarted an attack on an underwater cable linking Hawaii and the Pacific Region. This attack was enabled by a credentials-related breach of a third-party.

Cyber Attack on Undersea Cables

- **Incident involving disruption of internet connectivity due to compromised undersea cables.**
- **Consequences for global communication and commerce.**

Cybersecurity Challenges in Underwater Systems

Complex Environment:

- Difficulties in monitoring and securing underwater assets due to harsh and remote conditions.

Legacy Systems:

- Many underwater systems rely on outdated technology, making them vulnerable to cyber-attacks.

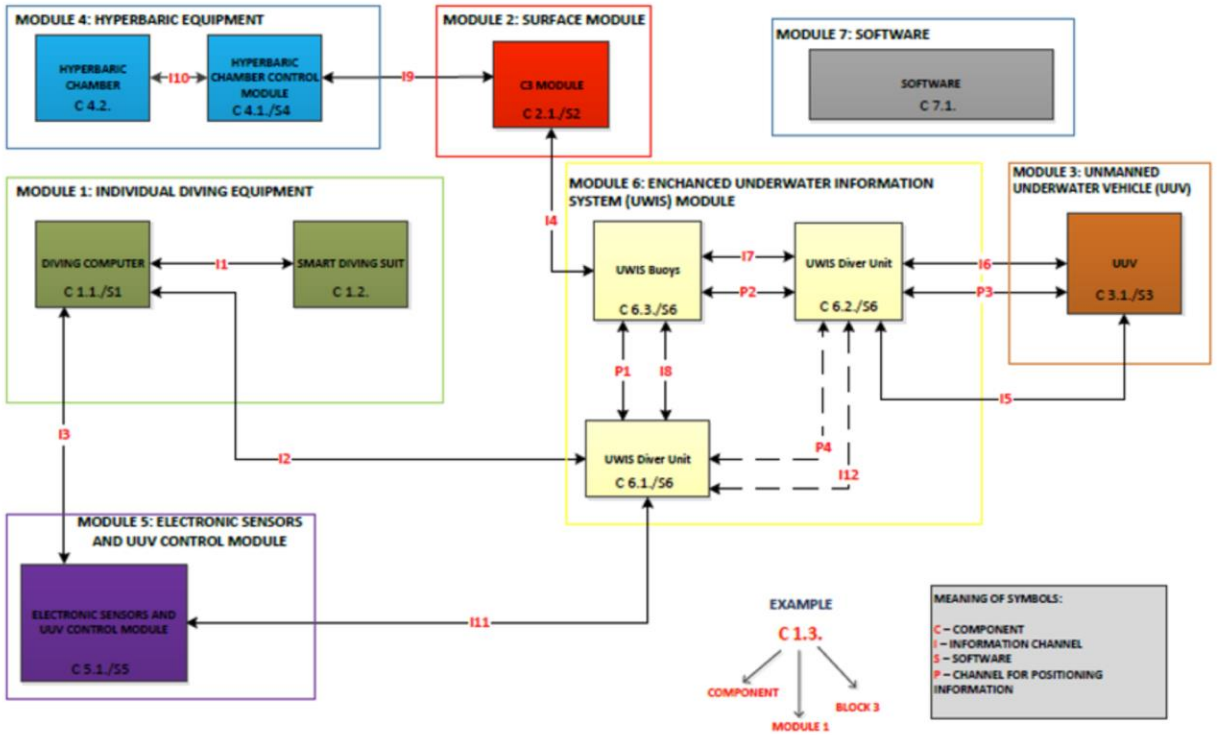
Integration with Modern Networks:

- Increasing interconnectivity with surface and land-based networks increases the attack surface.

CUIIS

Comprehensive Underwater Intervention Information System

BASIC DIAGRAM

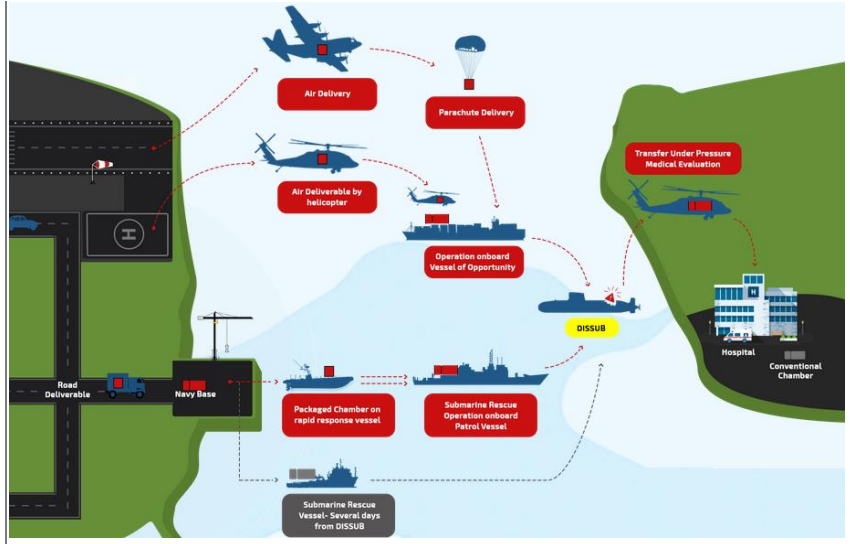


Underwater Traffic Control

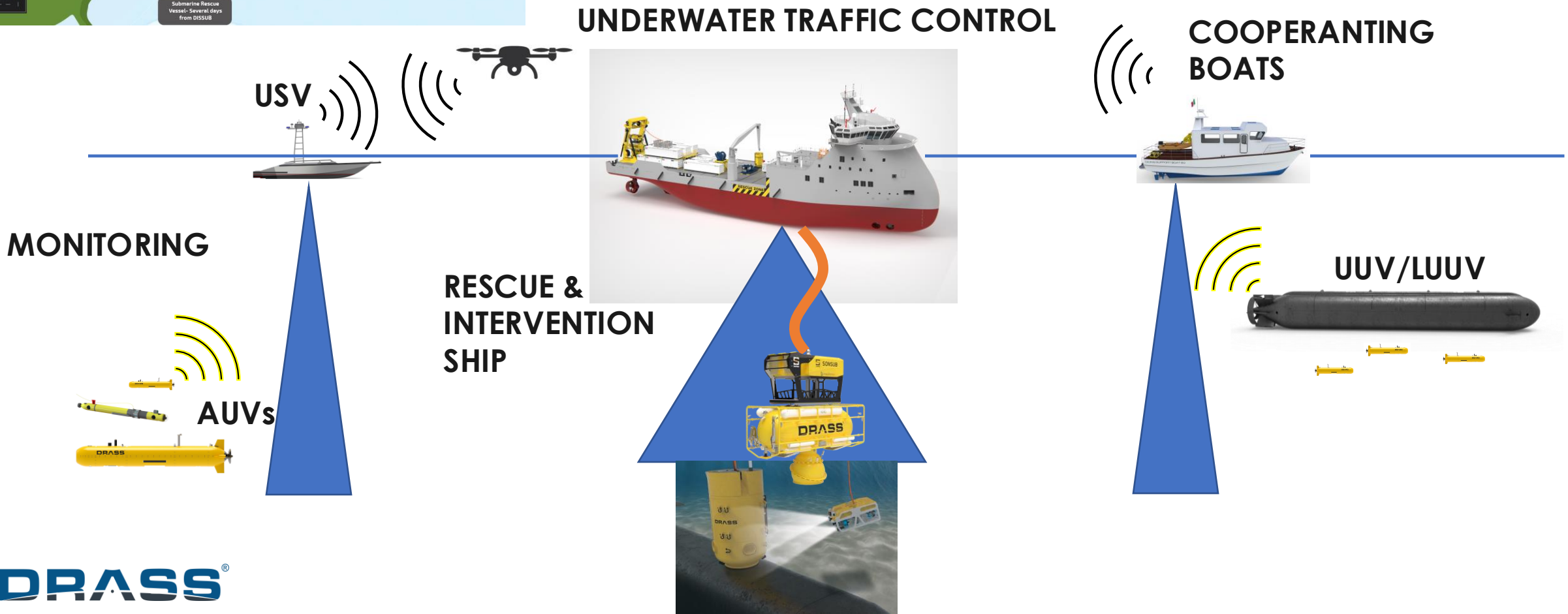
COM AEREE

COM SUB

COM CABLATE



The system is suitable for collecting information from delocalized scenarios and sharing it on the ground.



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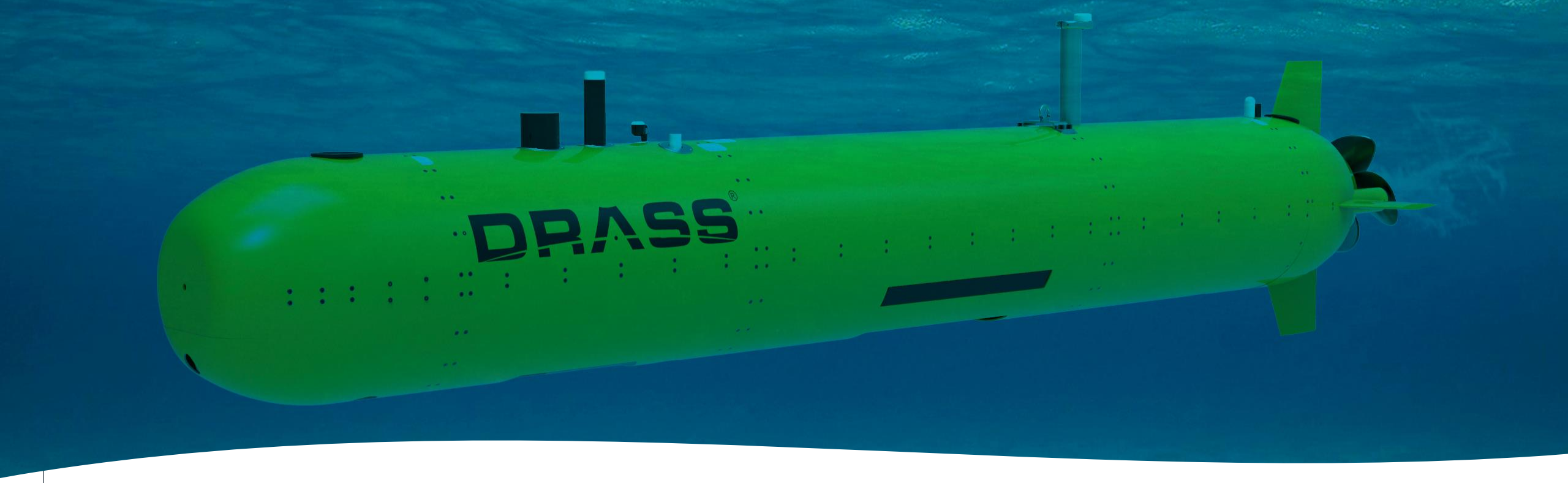


Defense Sector Strategies

Encryption and Secure Communication:

- Use of advanced encryption techniques to secure communication between underwater assets.

Network Segmentation:

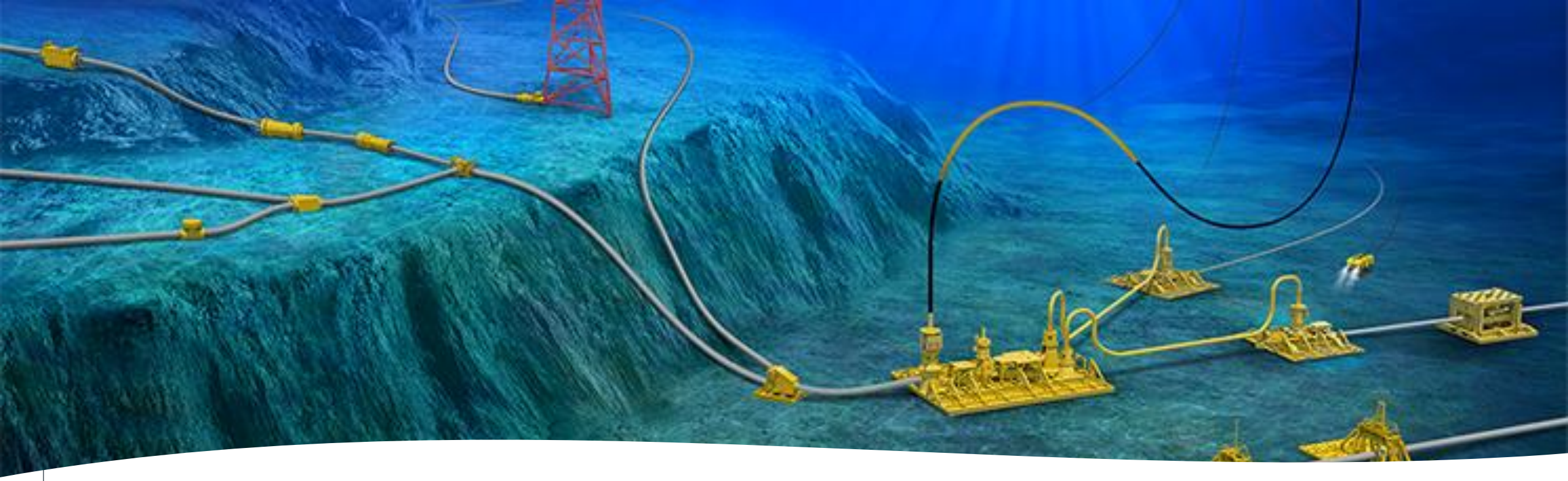
- Isolating critical systems to minimize the impact of a breach.

Intrusion Detection Systems (IDS):

- Deploying specialized IDS for underwater networks to detect and mitigate threats in real-time.

Redundancy and Backup Systems:

- - Implementing redundant systems and regular backups to ensure continuity of operations.



Commercial Sector Protection Strategies

Collaboration with Defense Sector:

- Sharing threat intelligence and cybersecurity practices between defense and commercial sectors.

Regular Audits and Assessments (health check):

- Conducting frequent security audits and vulnerability assessments on underwater assets.

Investment in R&D:

- Funding research into advanced cybersecurity solutions tailored for the underwater environment.

Public-Private Partnerships:

- Enhancing cooperation between governments and private companies to protect critical infrastructure.

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Future Trends

AI and Machine Learning:

- Use of AI for predictive threat analysis and autonomous defense mechanisms in underwater systems.

Blockchain for Secure Communication:

- Exploration of blockchain technology to ensure data integrity and secure communications.

Quantum Cryptography:

- Application of quantum cryptography for ultra-secure underwater communication.

Conclusion

Summary:

- Cybersecurity is a critical concern for both defense and commercial underwater systems.
- The increasing sophistication of cyber threats requires innovative and robust security measures.
- Collaboration, investment in technology, and continuous vigilance are key to safeguarding underwater assets.

Call to Action:

- Urge stakeholders to prioritize cybersecurity in underwater operations.
- Encourage ongoing research and development in this specialized field.

FP03-02-E34

DRASS®
UNDERWATERTECHNOLOGY

THANKS FOR YOUR ATTENTION

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